Environmental Noise Modelling in Hanoi : A Comparative Review of Old and New Urban Fabrics

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June 2026

Abstract

Rapid urbanization has elevated environmental traffic noise into a public health crisis linked to acute stress and chronic diseases. This survey reviews global traffic noise regulations (III) and evaluates its physical and psychophysiological health impacts (IV). By examining current noise analysis methodologies (V), we demonstrate that conventional Western frameworks, such as CNOSSOS-EU, require scarce high-resolution data and fail to accurately simulate developing megacities with motorcycle-dominated fleets. Furthermore, while emerging spatiotemporal deep learning models offer dynamic, data-driven alternatives, we highlight a critical gap in the literature: it remains unknown whether these models, currently validated only in European car-centric cities, can generalize to motorcycle-heavy soundscapes. To contextualize this modeling challenge, we provide a comparative analysis of Hanoi's contrasting urban fabrics (VI), highlighting how the Old Quarter's dense street canyons amplify noise through facade reflection, in stark contrast to the shielding patterns of modern high-rise districts. We conclude (VII) that motorcycle-dependent cities require localized acoustic mitigation and emphasize the critical need to train and validate modern predictive models on local acoustic signatures.

Index Terms — **Noise modelling, Hanoi, Public Health, Traffic noise, Soundscape, Smart City.**

I Introduction

Environmental noise is a pervasive pollutant that goes beyond pure decibel levels, acting as an environmental stressor that triggers immediate autonomic nervous system responses and is linked to the long-term development of chronic somatic diseases [2]. To effectively mitigate these psychophysiological impacts, urban planners rely on accurate road traffic noise modelling. However, state-of-the-art acoustic assessment frameworks require highly detailed traffic data, in cities with limited monitoring infrastructure, this input is often unreliable, which significantly reduces the accuracy of the resulting noise assessments [13]. Extrapolating these models to a developing megacity like Hanoi introduces further complexities. Traditional prediction methods were largely developed in Western contexts and often require significant modification to accurately capture Southeast Asia's unique traffic composition, which is heavily dominated by motorcycles and mopeds [6]. Furthermore, Hanoi presents a stark juxtaposition of urban morphologies. To address these modelling challenges, this paper presents a comparative review of traffic noise modelling in Hanoi, evaluating the distinct acoustic behaviors between its narrow, highly dense old quarters and its open, rapidly developing new urban fabrics.

II Review Methodology

A. Scope and Search Strategy

A systematic exploration of major scientific databases (e.g., Google Scholar, IEEE Xplore), alongside institutional and regulatory repositories such as EUR-Lex and WHO databases, was conducted. The search strategy employed combinations of keywords grouped into three primary domains: (1) *Acoustic Modelling and Artificial Intelligence* ("traffic noise", "acoustic sensing"); (2) *Health and Perception* ("psychoacoustic indicators", "psychophysiological stress", "dynamic exposure"); and (3) *Urban Dynamics* ("Hanoi motorcycle dynamics", "urban fabrics"). The theoretical data extracted from these peer-reviewed publications were subsequently cross-referenced with institutional policy frameworks.

B. Inclusion and Exclusion Criteria

To ensure the relevance and novelty of the survey, specific filtering criteria were applied to the literature search:

- **Inclusion Criteria:** The academic literature was focused on publications from the last ten years to capture the recent paradigm shift towards perception-based acoustics and data-driven models. The corpus includes studies addressing spatiotemporal deep learning, microscopic traffic simulation, low-cost acoustic sensor networks, and dynamic exposure routing. Furthermore, official regulatory documents, such as the World Health Organization guidelines, the European Noise Directive, and the Vietnamese QCVN 26:2025/BNNMT standards, were strictly included to ground the theoretical models in legislative reality.
- **Exclusion Criteria:** Studies focusing exclusively on indoor acoustics were excluded.

C. Corpus Composition

The final selected corpus synthesizes diverse yet complementary facets of environmental acoustics. The reviewed literature can be categorized into four core thematic groups:

1. **Health Impacts, Perception, and Dynamic Exposure:** Literature establishing the psychophysiological stress responses to noise, the necessity of psychoacoustic indicators beyond pure decibel levels, and the assessment of journey-time (dynamic) exposure to urban pollutants.
2. **Advanced Predictive Methodologies and AI:** Studies demonstrating the integration of Machine Learning (such as LSTM networks and gradient-boosted decision trees) and acoustic sensor networks to create surrogate models.
3. **Spatial Dynamics and WebGIS Integration:** Research focusing on spatial propagation within different urban fabrics, agent-based land use models evaluating building shielding effects, and the use of open geospatial data (WebGIS) to simulate traffic-noise exposure scenarios.
4. **Regulatory Frameworks:** Institutional guidelines and directives detailing exposure thresholds, health recommendations, and strategic noise mapping protocols.

III Regulations Review

A. International Health Guidelines

At the global level, the World Health Organization (WHO) provides foundational guidelines based on public health outcomes. Historically, WHO (2009) data indicated that over 43% of the European urban population was exposed to road traffic noise exceeding the critical threshold of 55 dB(A) [7]. Acknowledging the accumulating medical evidence regarding psychophysiological impacts, the WHO tightened these guidelines in 2018. Current recommendations urge public policies to limit road traffic noise exposure to 53 dB(A) for the L_{den} (day-evening-night) metric and 45 dB(A) for the L_{night} metric [2]. Furthermore, mitigation measures are officially recommended whenever the proportion of "highly annoyed" individuals within a population exceeds 10% [2].

B. Regional Frameworks: The European Union

Regional legislation provides actionable mandates based on these health guidelines. In Europe, the primary legislative driver is the Environmental Noise Directive (END, Directive 2002/49/EC), which requires Member States to produce strategic noise maps every five years for major transport axes and agglomerations exceeding 100,000 inhabitants, alongside action plans to manage and mitigate exposure [2, 5, 6, 13]. To ensure cross-border comparability, Directive (EU) 2015/996 established the CNOSSOS-EU framework, creating a mandatory, standardized methodology for noise assessment and modelling across the continent [2]. Additionally, in a broader approach to urban acoustics, Directive 2000/14/EC aims to harmonize acoustic emission limits for various outdoor equipment and machinery [6].

C. Local Legislation: Vietnam

In Vietnam, environmental noise is currently regulated by the National Technical Regulation on Noise, officially designated as QCVN 26:2010/BTNMT [8]. Promulgated in 2010, this directive superseded the older TCVN 5949:1998 standard [9]. The current regulation categorizes urban spaces into two primary zones based on their sensitivity. "Special areas," which include hospitals, libraries, kindergartens, schools, and places of worship, are subject to strict limits: 55 dBA during the day (06:00 to 21:00) and 45 dBA at night (21:00 to 06:00). In contrast, "Normal areas," comprising residential neighborhoods, hotels, and administrative offices, have a higher tolerance, with maximum permissible equivalent sound levels set at 70 dBA for daytime and 55 dBA for nighttime.

D. Technical and Methodological Standards (ISO)

The assessment of noise and its impact relies heavily on standardized international protocols. Subjective noise annoyance is formally quantified using specific socio-acoustic questionnaires governed by the ISO/TS 15666:2003 standard [2]. Furthermore, modern acoustic research increasingly integrates the ISO 12913 series, which shifts the paradigm from purely physical decibel measurements to the holistic evaluation of "soundscapes." Specifically, ISO 12913-1 defines the conceptual framework, ISO 12913-2 outlines in-situ data collection (e.g., via citizen-science initiatives), and ISO/TS 12913-3 specifies the data analysis protocols required to cross-reference human perception with acoustic measurements [2].

On the physical measurement side, the ISO 11819-2:2017 standard defines the Close ProXimity (CPX) method, which employs near-field microphones placed directly next to test tires at constant speeds (e.g., 20, 30, and 50 km/h) to isolate tire/road rolling noise [2]. Finally, to ensure the validity and international comparability of these vehicle noise emission tests, the ISO 10844:2021 standard dictates strict physical criteria (e.g., maximum aggregate size and asphalt content) for the construction of reference road surfaces [2].

To provide a comprehensive acoustic evaluation, additional ISO standards are routinely employed. The ISO 11819-1 standard outlines the Statistical Pass-By (SPB) method for measuring road traffic noise from individual vehicle pass-bys, complemented by ISO/TS 11819-3, which specifies the reference tires used for acoustic testing [2]. Environmental noise measurement outdoors is guided by ISO 1996-2 [2]. Regarding environmental adjustments, ISO/TS 13471-2 standardizes temperature corrections during noise measurements, while the ISO 13473 series (e.g., ISO 13473-5) is applied for determining pavement megatexture [2]. Lastly, to interpret the physical measurements perceptually, ISO 532-1 provides a standardized method for calculating loudness [2], and ISO 226 defines the normal equal-loudness contours that model human hearing sensitivity [5].

IV Global State of Noise pollution

A. Urban Trends and Demographic Pressures

Rapid urbanization and city densification have directly increased traffic volumes, transforming environmental noise into a severe public health threat [7, 5]. Urban morphology significantly exacerbates this exposure; for instance, dense street "canyons" amplify local noise through acoustic reflection, whereas strategies like fleet electrification yield different noise reduction impacts in dense districts compared to high-speed roads [3].

B. Key Hotspots and Vehicle Fleet Disparities

While noise is a global issue, major cities in developing nations have emerged as extreme noise pollution hotspots due to high population densities and limited financial resources for mitigation [6]. A critical distinction lies in vehicle fleet composition: unlike Western cities, which are predominantly car-centric, South and Southeast Asian megacities are dominated by motorcycles and mopeds [6]. These two-wheelers emit higher sound pressure levels than standard passenger cars [6]. Consequently, Western noise prediction models, which initially ignored motorcycles, must be significantly adapted to accurately reflect local acoustic realities in Asian cities, as previously demonstrated in studies from Taiwan [6].

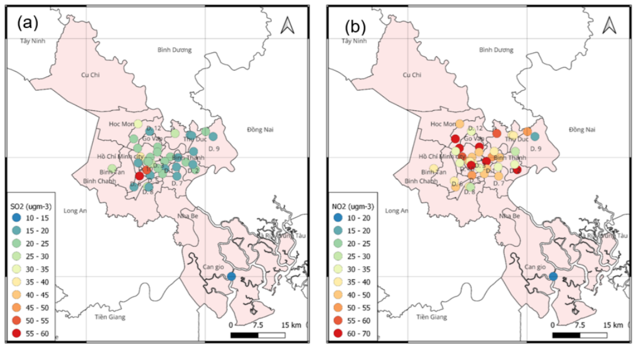


Figure 1: Map of pollutant concentration distribution, demonstrating high pollution accumulation along high-traffic networks in dense Vietnamese urban contexts [4].

C. Physical and Psychophysiological Health Impacts

Prolonged exposure to traffic noise exceeding 53 dB(A) is correlated with severe non-auditory health consequences, including cardiovascular diseases, hypertension, ischemic heart disease, and depression [2, 14, 3, 7, 5, 6]. In Europe alone, environmental noise accounts for the loss of over 1.5 million healthy life years annually [2]. In the short term, traffic noise triggers an involuntary psychophysiological stress response, altering autonomic nervous system biomarkers such as heart rate, blood pressure, and skin conductance [2, 5, 6].

Furthermore, road noise severely disrupts rest, leading to sleep disturbances, insomnia, and increased morning fatigue [2, 14, 3, 7, 6]. Extremely noisy environments can also lead to direct auditory damage, such as hearing loss and tinnitus, contributing to the global burden of over 1.5 billion people experiencing hearing decline [14, 7, 6]. Cognitive performance is also impaired, with noticeable reductions in memory and concentration, particularly in children [2, 14, 6]. Finally, the subjective perception of noise is quantified through "annoyance," with the percentage of "highly annoyed" individuals serving as a primary alarm signal for health risks [2, 7, 5]. Short, repetitive noise events exceeding 70 dB(A) are particularly notorious for generating high levels of subjective annoyance [6].

V Noise analysis methodologies

A. Continuous Metrics (L_{eq}, L_{den})

- **Definition and Role of L_{eq}:** The Equivalent Continuous Sound Level (L_{eq}) is a metric representing the average energy of a fluctuating sound over a specified period. Its primary role is to provide a single value containing the same overall acoustic energy as the actual sound pressure variations observed during the measurement timeframe [3, 6, 7].

- **Definition and Role of L_{den}:** The Day-Evening-Night Level (L_{den}) is an equivalent continuous noise level indicator that covers the day, evening, and night periods. It is typically calculated over a full year and serves as the standard metric for strategic noise mapping [2, 6].

- L_{den} **Penalty System:** To account for the heightened annoyance caused by noise outside of standard working hours, the L_{den} calculation system applies strict penalties: a 5 dB(A) weighting is added to evening hours (typically 18:00 to 22:00), and a heavier 10 dB(A) penalty is applied to nighttime hours (22:00 to 06:00) [6, 7].
- **Limitations of Continuous Metrics (The "Smoothing" Effect):** Energy-averaged metrics such as L_{den} and L_{eq} highly compress acoustic data, effectively smoothing out distinct sound events. Consequently, they tend to mask or drown out transient and impulsive noise events within the overall average. Because they lose crucial information regarding sudden temporal fluctuations and the spectral characteristics of the acoustic environment, these metrics poorly describe the true annoyance levels perceived by local populations [2, 5].

B. Event-Based Metrics (L_{max})

- **Definition of L_{max}:** The Maximum Sound Level (L_{max}) represents the highest A-weighted sound pressure level (the noise peak) recorded during a vehicle pass-by or a specific time interval [2, 6].
- **Importance for Acute and Sudden Noises:** L_{max} is a crucial metric for overcoming the shortcomings of energy averages. Short, sudden acoustic events (such as horn honking or acceleration peaks) generate a high degree of subjective annoyance. Studies indicate that repeated impulsive noises cause severe annoyance whenever their maximum intensity repeatedly exceeds the 70 dB(A) threshold throughout the day [6, 7].
- **Link to Motorcycle Dynamics:** Unlike the continuous rolling noise generated by cars, traffic composed heavily of motorized two-wheelers is characterized by highly intense sound events. Motorcycles and mopeds emit significantly higher maximum sound pressure levels (L_{max}) than standard passenger cars, particularly during hard accelerations at low RPMs, where engine noise and acoustic roughness heavily dominate the soundscape [2, 6]. In the specific context of Hanoi, historical measurements have demonstrated that transient events like heavy horn usage can cause acute noise spikes reaching 90 to 100 dB(A) [10].

C. Framework Comparison: European vs. Asian Modelling

- **Characteristics of European Models (e.g., CNOSSOS-EU):** Western modelling frameworks, such as CNOSSOS-EU, were designed to harmonize noise prediction across EU Member States. Inherently car-centric, these models primarily distinguish between light vehicles and heavy-duty trucks. Crucially, they require high-resolution input data regarding traffic flows and vehicle speeds to function accurately [6, 7, 13].
- **Limitations in Southeast Asian Megacities:** The application of these Western models is severely limited in developing Asian megacities. First, the vehicle fleet in regions like Southeast Asia (and Taiwan) is overwhelmingly dominated by two-wheelers (motorcycles and mopeds) a category that initial Western models, such as the Nordic prediction method, entirely ignored. Second, these densely populated regions often lack expensive traffic monitoring infrastructure (e.g., radar systems), rendering the high-resolution input data required by complex models like CNOSSOS-EU either non-existent or highly unreliable [6, 13].
- **Necessary Alternatives and Adaptations:** To accurately model Asian urban noise, two major adaptations are required. Methodologically, existing prediction equations must be modified to integrate the specific emission profiles of motorized two-wheeler fleets, an approach successfully demonstrated by Taiwanese researchers adapting the Nordic model [6]. On the data collection side, to bypass the lack of cameras and radars, an innovative alternative involves deploying low-cost noise sensor networks. These networks act as surrogate traffic sensors, indirectly estimating vehicle flows based on acoustic signatures to continuously feed spatial prediction models [13].

D. Software Tools and Field Measurement Best Practices

- **Field Measurement Best Practices:** Accurate noise assessment relies on strict adherence to empirical field measurement protocols, primarily governed by standards such as ISO 1996-2 [2]. Best practices dictate that measuring equipment must be carefully positioned, typically at a standard height of 1.2 to 1.5 meters above the ground, and placed at a minimum distance from reflective surfaces (e.g., building facades) to avoid acoustic interference. Furthermore, meteorological variables (wind speed, humidity, and temperature) must be continuously monitored and mathematically corrected to ensure data integrity [2].
- **Software Tools for Noise Mapping:** Once empirical data is collected, spatial acoustic propagation is simulated using advanced software tools. While commercial packages such as SoundPLAN, IMMI, or CadnaA dominate traditional physics-based modelling (e.g., implementing the CNOSSOS-EU framework), there is a growing trend toward integrating open-source geospatial tools (such as QGIS and WebGIS) for urban planning [5]. Recently, researchers have also begun developing surrogate models using

Python-based machine learning libraries to bypass the heavy computational costs of traditional software, allowing for dynamic, data-driven exploratory analysis [3].

VI Comparative Analysis of Hanoi's Urban Fabrics and Traffic Dynamics

A. Historical Baseline of Hanoi's Traffic Dynamics

To understand the current acoustic challenges in Hanoi, it is essential to establish a historical baseline. Environmental monitoring conducted between 1996 and 2005 reveals that Hanoi experienced an explosive four-fold increase in vehicle flow [10]. During peak commuting hours, motorized two-wheelers constituted up to 91% of the total traffic volume [10].

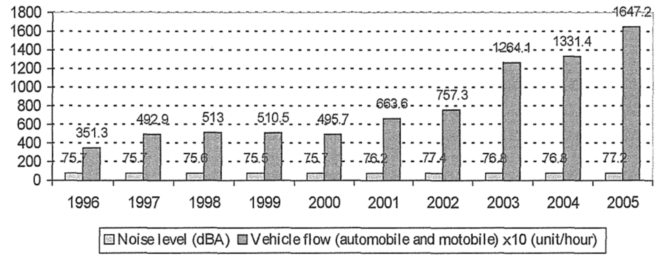
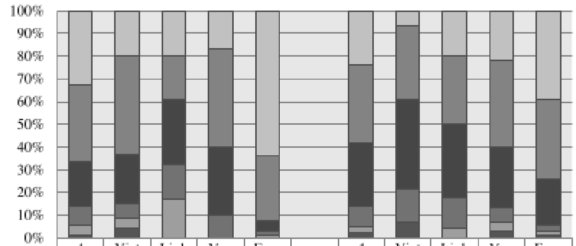


Figure 2: The change of daily average of road traffic noise levels and vehicle flows at National road 1A in the Southern of Hanoi from 1996 to 2005 [10].

This unique vehicular composition fundamentally alters the urban soundscape. For instance, continuous horn usage within this dense motorcycle stream generates acute maximum noise levels (L_{max}) reaching between 90 and 100 dB(A) [10]. Interestingly, despite a massive 4.7-fold increase in traffic volume on major arteries over this decade, the average equivalent noise levels only increased by 0.5 to 2.8 dB(A), a counterintuitive phenomenon attributed to concurrent improvements in road surface quality and wider infrastructures [10].

B. Contrasting Urban Morphologies: Old Fabrics vs. KDTM

The ancient center of Hanoi, stemming from spontaneous spatial development, is characterized by an ingrained "culture of alleys" [12]. This historical urban fabric features narrow, winding streets and dead-ends, often lacking sidewalks, which host an extremely high population density and mixed-use spaces where commercial activities occupy the front of traditional tube houses [12]. In stark contrast, "Khu Do Thi Moi" (KDTM) represent large-scale, planned new residential areas that emerged in the late 1990s to modernize housing [12]. Unlike the old alleys, KDTM architecture relies on excessively wide roads surrounding modern apartment blocks [12]. From their inception, these spatial models were designed to facilitate automobile access and parking, rendering them inherently car-centric environments [12].



	4 KDTM	Viet Hung	Linh Dam	Van Quan	Eco-park		4 KDTM	Viet Hung	Linh Dam	Van Quan	Eco-park
Very satisfied	32.6	19.6	19.6	16.7	63.8		23.5	6.5	19.6	21.7	39.1
Satisfied	33.9	43.5	19.6	43.3	29.0		34.4	32.6	30.4	38.3	34.8
Quite satisfied	19.9	21.7	28.3	30.0	4.4		28.5	39.1	32.6	26.7	20.4
Unsatisfied	7.7	6.5	15.2	10.0	1.4		8.6	15.2	13.0	6.7	2.9
Very unsatisfied	5.0	4.3	17.3	0.0	1.4		2.3	0.0	4.4	3.3	1.4
No answer	0.9	4.4	0.0	0.0	0.0		2.7	6.6	0.0	3.3	1.4

Figure 3: Assessment of the satisfaction of KDTM residents regarding inner-KDTM traffic and external connectivity to the city [12].

Urban morphology directly influences acoustic propagation. Dense environments bordered by high-rise buildings form "street canyons" that trap sound waves and amplify local noise exposure through facade reflections [3]. From a traffic perspective, the spatial layout of KDTMs generates significant negative externalities. Because these modern districts often lack fully developed internal services, residents must commute en masse to the old center for work or education. This generates severe traffic bottlenecks and intense noise pollution at the entrances and exits of these newly built areas during peak hours [12].

C. Traffic Dynamics, Noise, and Livability

To assess the human impact of these dynamics, researchers replicated Donald Appleyard's seminal 1969 "Livable Streets" methodology in Hanoi, conducting field experiments to compare the perceived livability of residents living on high-traffic versus low-traffic streets [11]. The findings indicate that intense traffic significantly degrades residents' quality of life. High-volume streets are perceived as highly dangerous, characterized by inaccessible sidewalks and aggressive drivers, forcing parents to forbid their children from playing outdoors [11]. Furthermore, this traffic generates chronic stress through severe air and noise pollution. Residents on noisy arteries report frequent disturbances while sleeping, eating, or conversing, often isolating themselves by keeping windows permanently closed [11]. The elderly are particularly vulnerable, with 40% expressing fear of walking on high-traffic streets, compared to only 22% in quieter zones [11].

However, Hanoi presents a surprising contextual nuance regarding social interactions. Unlike Appleyard's original Western findings, which observed a drastic reduction in neighborly relations due to traffic, the Hanoi study reveals that social interaction, alongside feelings of privacy and home territory, remains high across all streets regardless of traffic and noise levels [11]. This resilience is attributed to Vietnam's strong collective culture, a robust social fabric, mixed-use street activities, and the fact that families often reside in the same house for several decades [11].

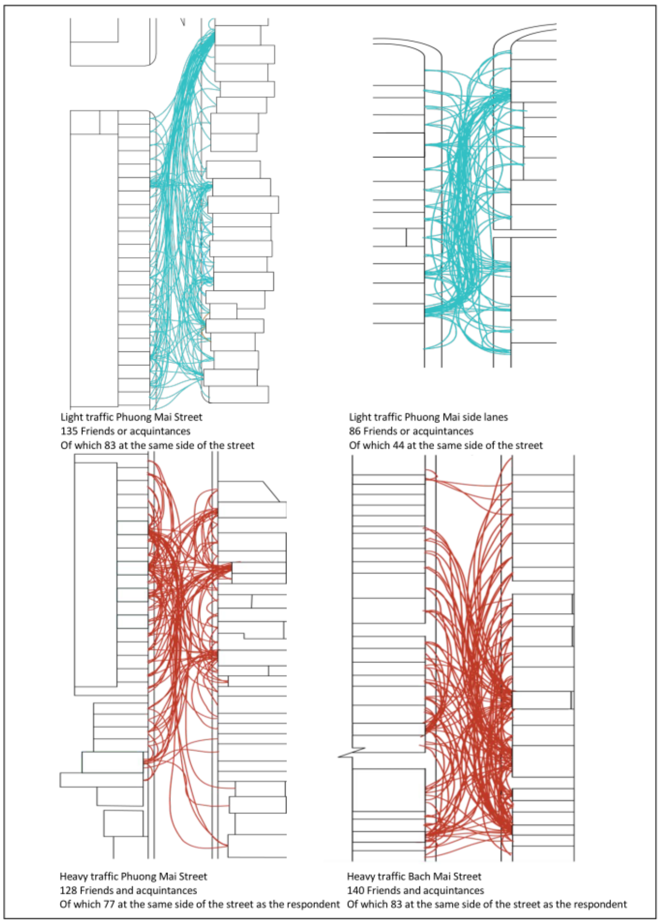


Figure 4: Composite maps indicating social interactions and friendships along Phuong Mai and Bach Mai streets, demonstrating resilient social networks despite traffic volumes [11].

D. Policy Tensions and the 2030 Motorcycle Ban

In response to escalating accidents, severe congestion, and environmental degradation caused by exhaust fumes and continuous horn honking, the Hanoi City Council approved a radical plan in 2017 to restrict and eventually impose a total ban on motorcycles in central districts by 2030 [12].

However, this motorcycle eradication policy raises immense paradoxes. Historically, the motorcycle is the most deeply rooted transport mode in Vietnamese daily life, primarily because it is the only vehicle genuinely suited to navigate the narrow morphology of the ancient alleys [12]. It also remains an indispensable economic tool for the working class and the informal economy, facilitating deliveries and motorbike-taxis [12].

Consequently, the ban risks forcing a transition toward a purely car-centric city. This crystallizes the tension between the authorities' vision of a modern, two-wheeler-free metropolis and the demographic realities of Hanoi [12]. Projections suggest that by 2030, public transport will only be capable of meeting 50–55% of mobility demands [12]. By banning motorcycles without curbing the massive surge in private car ownership, which is widely considered a symbol of success among KDTM residents, authorities risk replacing motorcycle traffic jams with insurmountable automobile congestion, ultimately failing to address the core challenges of sustainable urban mobility [12].

VII Conclusion

This survey has demonstrated that environmental traffic noise in Hanoi is not merely a byproduct of urbanization, but a critical public health challenge shaped by the city's unique morphological and demographic realities. Traditional Western acoustic frameworks, such as CNOSSOS-EU, alongside energy-averaged continuous metrics (L_{eq}, L_{den}), are fundamentally unequipped to accurately model Southeast Asian megacities. They consistently fail to account for the transient, high-intensity noise spikes (L_{max}) generated by motorcycle-dominated fleets and pervasive horn usage, thereby underestimating the true psychophysiological stress imposed on residents.

Furthermore, Hanoi's acoustic landscape is complicated by the stark juxtaposition of its historical "culture of alleys" and the rapid emergence of modern, car-centric new residential areas (KDTM). While the dense street canyons of the Old Quarter trap and amplify noise through facade reflections, the functional disconnect of KDTMs forces mass commuting, shifting and intensifying traffic bottlenecks across the city. In this context, the municipality's proposed 2030 motorcycle ban appears as a potentially counterproductive policy. Without a comprehensively developed public transit network, eradicating motorcycles risks replacing two-wheeler noise with insurmountable automobile congestion, entirely neglecting the spatial constraints of the old urban fabric and the socio-economic reliance of the population on motorized two-wheelers.

To initiate the necessary paradigm shift toward localized, data-driven methodologies, our subsequent research phase will focus on empirical data collection across three contrasting urban typologies: the highly dense Old Quarter (Hoan Kiem), an emerging smart community characterized by heavy construction (Ocean Park), and a major high-speed transport infrastructure corridor (Vinh Tuy). By employing a calibrated network of professional acoustic sensors and pervasive smartphone-based measurements, we aim to capture high-resolution temporal noise profiles. These localized datasets will subsequently be integrated into an agent-based modeling environment (GAMA Platform) to simulate dynamic noise propagation and evaluate specific public health impact indicators. Ultimately, this predictive simulation framework will enable the robust assessment of various noise mitigation scenarios, providing actionable guidelines for sustainable urban planning in Hanoi.

Acknowledgments

The authors would like to express their sincere gratitude to the College of Engineering and Computer Science at VinUniversity for providing the academic environment and resources that facilitated this research. We also extend our appreciation to our professors and academic advisors for their guidance and support throughout the development of this comparative review. The authors would like to acknowledge the use of NotebookLM to assist in structuring the major sections of this survey paper and cross-referencing the selected source materials.

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